

Acceptance Decision and Reviews of Submission 26 for MECC 2026 (July 2, 2026)

Acceptance Decision and Reviews of Submission 26 for MECC 2026	
Submission number	26
Authors or proposers	Sesha Charla*,Peter H. Meckl,Bin Yao
Title	Modelling and Parameter Estimation of Diesel Engine Aftertreatment System under Catalyst Saturation for Aging Diagnostics
Scroll down to view the publication decision and reviews when and if available	

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Publication Decision for this Submission

Timestamp	June 26, 2026 07:05:56
Decision	Accepted as Joint MECC-Journal paper submission
Cover message	Dear Mr. Sesha Charla, We are pleased to inform you that your Joint MECC-JDSMC Submissions paper number 26, titled "Modelling and Parameter Estimation of Diesel Engine Aftertreatment System under Catalyst Saturation for Aging Diagnostics", has been accepted for inclusion in the 2026 Modeling, Estimation and Control Conference (MECC 2026). MECC 2026 is sponsored by the American Automatic Control Council (AACC), and co-sponsored by the International Federation of Automatic Control (IFAC). The conference will be held on October 25-28, 2026 in Phoenix, AZ, USA. Your paper was reviewed by peers from the modeling, estimation, and control community, recruited by associate editors from the Conference Editorial Board (CEB). MECC 2026 acceptance/rejection decisions were made based on criteria reviewed and approved by the conference's International Program Committee (IPC). Reviewer and associate editor comments are available through PaperCept (ifac.papercept.net). Please take these comments into account as you prepare your final manuscript for submission. The journal decision is appended at the end of this notification. You will NOT receive a separate decision letter from the journal EiC. Based on the journal EiC decision, your paper will be moved to the "Joint MECC-Journal presentation only abstract" category, and you need to submit an 1-page final Abstract rather than the full paper by July 20, 2026, through PaperCept. At least one of the authors is still expected to attend MECC26 and present the paper at the conference. Please note that at least one author is required to pay full conference registration in order for final abstract submission of MECC 2026. Your final submission must comply with IFAC formatting guidelines. Please visit the "Authors" page of the MECC 2026 website (mecc2026.a2c2.org) for IFAC paper templates. The authors take responsibility for the material that has been submitted. IFAC-sponsored conferences will abide by the highest standard of ethical behavior in the review process as explained on the Elsevier webpage (https://www.elsevier.com/authors/journal-authors/policies-and-ethics), and the authors will abide by the IFAC publication ethics guidelines (https://www.ifac-control.org/events/organizersguide/PublicationEthicsGuidelines.pdf/view). The detailed MECC 2026 technical program, including the presentation time slot for your paper and its assignment to a contributed or invited session (in case of invited session papers), will be announced by early September 2026. Conference registration will open within the next two weeks. We have an exciting program for MECC 2026 including

contributed sessions, invited sessions, plenary talks, special sessions, tutorial sessions, and workshops of interest to the modeling, estimation, and control community.

Thank you for helping us make MECC 2026 a success. We look forward to seeing you at the conference!

Sincerely Yours,

Juan Ren (Program Chair)
Manish Kumar (General Chair)
Jin-Oh Hahn (Editorial Board Chair)

Subject: Joint MECC26 – DSMC Potentially Acceptable – Minor Rev

Dear MECC26 Author:

I am pleased to inform you that your MECC26 paper has been provisionally accepted for publication in the ASME Journal of Dynamic Systems, Measurement and Control (JDSMC), pending minor revisions.

This recommendation was based on a thorough review of your joint submission paper, handled in most cases by an Associate Editor of the JDSMC. This recommendation guided my decision as Editor-in-Chief, and was reviewed by the MECC Editorial Board.

Although your paper has been provisionally accepted for JDSMC, at least one of the authors is still expected to attend MECC26 and present the paper at the conference. You need to send an 1-page Abstract to the conference rather than the full paper, per instructions included in your acceptance letter for the joint submissions to MECC 2026 & JDSMC. At least one of the authors is still expected to attend MECC26 and present the paper at the conference.

We ask that you address the issues raised by the reviewers and the AE and submit your revised manuscript and detailed rebuttal before July 19, 2026 (ideally by July 6), for expedited review at the ASME JDSMC site (https://journaltool.asme.org).

Your JDSMC journal submission (revised manuscript) should be formatted in two-column JDSMC style. You should highlight any changes you've made relative to the original MECC submission. Your detailed list of the revisions and rebuttal should be compiled and submitted in a combined pdf with your revised manuscript. The revised manuscript will be reviewed by an Associate Editor of the JDSMC, most likely the same one who managed your original manuscript, and me. If in doubt, we'll also ask one of the MECC reviewers to confirm the important revisions.

Instructions on accessing your reviews at the MECC26
- You can access your MECC26 paper reviews by going to the MECC26 conference website: https://ifac.papercept.net/conferences/scripts/start.pl
- Once you login to the MECC26 website, select "Enter" next to the "Author or proposer of submission" corresponding to your paper number.
- Click on "Choose an option" for your paper number and select "Inspect the reviews and decision letter."

Instruction on submitting your revised JDSMC manuscript
- First, select the journal: ASME Journal of Dynamic Systems, Measurement, and Control
- Then, Enter Paper Details.
- Use the link to upload your revised manuscript (two-column ASME Journal format), followed by your rebuttal in a single PDF.
- Please answer all the questions under Conference Acknowledgement:
o Has this paper been submitted, presented, or published at a conference? Yes
o Has permission to submit and copyright permission been obtained? Yes
o Was this an ASME Conference? Yes
o Then select 2026 and the MECC in the drop-down menu, and provide the full Paper Number (e.g., MECC26_XXX).

If you have any questions on this, email me (annastef@umich.edu) and cc: jds@publishingsolutions.net with the email subject "Joint MECC26_XXX" (the word "Joint" and your MECC26 paper number).

Please upload your revised paper to the JournalTool before July 19 and ideally by July 6 so that we can expedite the publication process during the summer vacation period and hopefully have your DOI number and your QR code before the MECC26 conference!

Thanks, Anna G. Stefanopoulou
Editor-in-Chief
ASME JSMC

Comments to the author

Comments to author (Associate Editor)
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The paper discusses an important and interesting research problem about modeling and parameter estimation of Diesel engine aftertreatment system with experimental validations.

Some technical details (e.g., justify the necessary condition of the proposed method based on sufficient data under vehicle working conditions) need to be clarified, and some writing issues (e.g., Appendix-?? on page 5) need to be revised.

Reviews of this Submission

Reviewer number	Review ID	Comments to the author	Attachment to the review

12	694	1. The significance of this paper should be stated clearly. 2. There are some writing mistakes in this paper. Such as: In page 5, the expression of "(Appendix-??)" is wrong. 3. The readers may want to see the clear comparison of experimental data and model results. It is better to present the figures but the numbers. 4. The literature review is not sufficient. The limitations of the existing works are not clear.	
14	1886	## Overall Evaluation This paper presents a novel approach for modeling and parameter estimation under catalyst saturation in a diesel engine SCR-ASC aftertreatment system to achieve onboard catalyst aging diagnostics. By utilizing the response under saturation as a system upper bound constraint, the study cleverly addresses the unidirectional bias uncertainty caused by the NO_x sensor's cross-sensitivity to ammonia. The authors provide thorough validation using both engine test-cell data and real-world heavy-duty truck road driving data, demonstrating that the proposed algorithm holds significant potential for practical engineering applications. The paper is well-structured, tightly integrating theoretical derivation with engineering validation. ## Detailed Comments 1. Formatting of Chemical Compounds: NO_x, NH_3, O_2, and N_2 must be strictly rendered in roman font. 2. Engineering Feasibility and Necessary Conditions: The authors state that a necessary condition for the proposed parameter estimation and aging detection framework is that the data must contain a "sufficiently long" or "sufficiently many" samples close to catalyst saturation. However, during real-world heavy-duty truck operations, the vehicle may encounter prolonged light-load cruising, extended downhill driving, or specific conservative driving habits, which might result in entire driving segments failing to trigger catalyst saturation at all. Under such scenarios, will the system generate false alarms for catalyst aging? In a practical onboard OBD strategy, how long is the system expected to wait, or under what specific operating conditions can it ensure this necessary condition is satisfied? It is recommended that a supplementary description or discussion regarding this be added to the main text. 3. Interpretation of Threshold β: In the test-cell validation, the decision thresholds are set to $\beta_{\text{hFTP}} = 4.00$ and $\beta_{\text{RMC}} = 10.00$, respectively. Why is the threshold abruptly scaled up directly to $\beta_{\text{truck}} = 250$ in the real-world truck driving segment analysis?	
18	2361	This paper presents a practical to diagnosing catalyst aging in diesel Selective Catalytic Reduction (SCR) systems. It proposes a model that explicitly accounts for the interplay between a gas parcel's residence time and the sensor's sampling frequency. The proposed works suffers from the limitation that the parameter estimation algorithm fundamentally requires the data to contain segments where the catalyst is fully saturated. As the authors acknowledge, if the catalyst does not reach saturation, the algorithm incorrectly fits the maximum observed surface coverage as the absolute capacity, which would entirely skew the aging diagnostics. This is a severe bottleneck for real-world deployment, evidenced by the fact that the actual truck drive segments evaluated in the study featured very few samples (often under 10%) near saturation. Furthermore, the authors note: "The saturated catalyst model’s response does not depend on the previous state but only on the inputs, making the response independent of the initial conditions." This effectively means (at least to my understanding) that the dynamics are irrelevant, and that is the reason a simple MLE point estimator suffices. In that sense, this paper does not fit the scope of MECC, because the system dynamics don't really matter.	

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